ADVANCING GLOBAL COMMUNICATIONS

TIA STANDARD

Project 25

Inter-RF Subsystem Interface Messages and Procedures for Voice Services, Mobility Management, and RFSS Capability Polling Services- Addendum 1- Group Emergency Behaviors

TIA-102.BACA-B-1 **July 2013**
(Addendum to TIA-102.BACA-B)

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Foreword

(This foreword is not part of this standard.)

This document has been created to further the objectives outlined in a Memorandum of Understanding (MOU) dated December, 1993 between APCO/NASTD/FED and the Telecommunications Industry Association (TIA).

The MOU provides that APCO/NASTD/FED will devise a Common System Standard for digital public safety communications (the Standard), and that TIA shall, to the extent TIA members are willing, provide technical assistance in the development of documentation for the Standard. This document has been developed by TR8.19 (Wire-line Interfaces) with inputs from the APCO Project 25 Interface Committee (APIC), the APIC ISSI Task Group, and TIA Industry members.

The appearance of the abbreviation “P25” or the phrase “Project 25” on the cover sheet of this document when published or in the title of any TIA-102 document or Telecommunications Service Bulletin (TSB) referenced herein, indicates the APCO/NASTD/FED Project 25 Steering Committee has adopted the document as part of the Standard. The appearance of the abbreviation “P25” or the phrase “Project 25” on the cover sheet of this document or in the title of any document referenced herein does not limit the applicability of the information contained in the document to “P25” or “Project 25” implementations exclusively.

This document is an Addendum to **[102BACA]**. The ISSI is the interface between Radio Frequency Subsystems (RFSSs) satisfying relevant user requirements as described in **[102BACC]**.

Patent Identification

The reader's attention is called to the possibility that compliance with this document may require the use of one or more inventions covered by patent rights. By publication of this document no position is taken with respect to the validity of those claims or any patent rights in connection therewith. The patent holders so far identified have, we believe, filed statements of willingness to grant licenses under those rights on reasonable and nondiscriminatory terms and conditions to applicants desiring to obtain such licenses.

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IMBE™ is a trademark of Digital Voice Systems, Inc.[1]

[1] Use of the trademark identified in this standard is not an endorsement by TIA.

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DOCUMENT REVISION HISTORY

Version	Date	Description
Rev 0	June, 2013	Approved for TIA publication.

A. Addendum Introduction

The purpose of this addendum is to update information contained in **[102BACA]**, to include procedural changes to support consistent group emergency behaviors between RFSSs that may implement different group emergency tracking behaviors per **[102AABD]**.

B. Addendum Scope

This addendum provides the following changes to the ISSI:
(1) Enhanced Group Emergency procedures

This document will only include information relevant to the aforesaid scope. The reader is referred to **[102BACA]** for all other pertinent information. The next complete revision of **[102BACA]** will incorporate the changes contained in this addendum. Care must be taken when incorporating changes in this addendum to avoid conflicts with the changes made by other addendum(s) that are incorporated at the same time.

Changes to **[102BACA]** are described in this document starting at Section 1.0. *[Italic text between square brackets provides instructions on how to incorporate changes in a section.]* If a section contains subsections, and neither the section nor subsections thereof change, only the section will be identified in this document. Note that section numbers identified in this addendum correspond to the same section numbers in **[102BACA]**. Also note that new or modified text is tracked in blue in order to assist the reader.

C. Addendum References

This addendum references the following TIA documents.

1. Normative References

[102BACA] *Project 25 Inter-RF Subsystem Interface Messages and Procedures for Voice Services, Mobility Management, and RFSS Capability Polling Services*, TIA-102.BACA-B, TIA, November 2012

[102BACC] *Project 25 Inter-RF Subsystem Interface Overview*, TIA-102.BACC-B, TIA, November 2011

[102BACD] *Project 25 Inter-RF Subsystem Interface - Messages and Procedures for Supplementary Data*, TIA-102.BACD-B, TIA, July 2011

[102AABD] *Trunking Procedures*, TIA-102.AABD-A, TIA, December 2008

[102AABC] *Project 25 Trunking Control Channel Messages*, TIA-102.AABC-C, TIA, October 2009

1. Introduction

*[No change from existing text in this section of **[102BACA]**, except for subsection 1.4 below.]*

1.1 Scope

*[No change from existing text in this section of **[102BACA]**.]*

1.2 Terminology

*[No change from existing text in this section of **[102BACA]**.]*

1.3 Document Structure

*[No change from existing text in this section of **[102BACA]**.]*

1.4 Definitions

*[Add the following new definitions in this section of **[102BACA]**.]*

Non-Tracking RFSS: This term is used to denote an RF Subsystem that does not track the emergency state of groups per **[102AABD]**. A non-tracking RFSS can consider some or all SUs “sufficiently privileged” with regards to the Emergency Cancel CAI messages.

Tracking RFSS: This term is used to denote an RF Subsystem that tracks the emergency state of groups per **[102AABD]**. A Tracking RFSS registers for Group Supplementary Data to ensure that Group Emergency Cancel messages are received.

1.5 Abbreviations

*[No change from existing text in this section of **[102BACA]**.]*

1.6 References

*[No change from existing text in this section of **[102BACA]**.]*

2. Overview of Architecture and Protocol Suite

[No change from existing text in this section of ***[102BACA]****.]*

2.1 Architecture Overview

[No change from existing text in this section of ***[102BACA]****, except for subsection 2.1.3.]*

2.1.1 Call Models

[No change from existing text in this section of ***[102BACA]****.]*

2.1.2 Subsystem Topologies

[No change from existing text in this section of ***[102BACA]****.]*

2.1.3 Functional Model

[Replace existing text in this section of ***[102BACA]*** *with the text below.]*

In this release, the ISSI supports only inter-working between RFSSs in trunked mode using the following functional services, illustrated in Figure 9:

1) Mobility Management
2) Call Control
3) Media Processing (Transmission Control)
4) RFSS Service Capability Polling

"Mobility Management" is a set of functional services corresponding to the registration, affiliation and authentication functions of the TIA-102 air interface. Mobility Management for a console corresponds to a console's group affiliation. Mobility management comprises two basic functional entities, the Unit Mobility Management Function (UMMF) and the Group Mobility Management Function (GMMF). An additional functional entity, the Packet Data Mobility Management Function (PDMMF), is required for TIA-102 Packet Data mobility management, as is detailed in Section 5 of this document. Mobility Management is provided via the SIP protocol.

"Call Control" is a set of functional services related to the control signaling for ISSI calls. It includes mechanisms for call setup, call teardown, management of RTP resources, determination of the allowed vocoder mode for talk spurts on an ISSI call segment, and the coordination of confirmed calls. Call Control also includes coordination of Group Emergency Cancellation. Call control comprises two functional entities, the Unit Call Control Function (UCCF) and the Group Call Control Function (GCCF). Call Control is also provided via the SIP protocol.

"Media Processing" (or "Transmission Control") is concerned with the management of talk spurts within an ISSI call. It includes mechanisms for arbitrating simultaneous or nearly simultaneous call requests, for the management of losing audio, and for the conveyance of voice information. When consoles participate in group calls, media processing also includes console priority which enables higher priority console transmission requests to take precedence over a talk spurt of a lower priority console transmission request. Media Processing comprises two functional entities, the Master Media Function (MMF) and the Subordinate Media Function (SMF). Which media function is active for a particular call is determined by the role of the RFSS in the call. Media processing is provided via RTP.

"RFSS Service Capability Polling" is an optional ISSI support function that allows an RFSS to:

- Determine if it can communicate with another RFSS connected using the ISSI.
- Obtain the other RFSS's service capability information if the other RFSS supports sending this information.
- Update the other RFSS with any change in the originating RFSS' service capability information.

It includes a mechanism for an RFSS to request service capability information from another ISSI-connected RFSS. RFSS Service Capability Polling comprises one functional entity, the RFSS Service Capability Polling Function (RCPF). RFSS Service Capability Polling is provided via the SIP protocol.

These functional services are described in detail in the following sections.

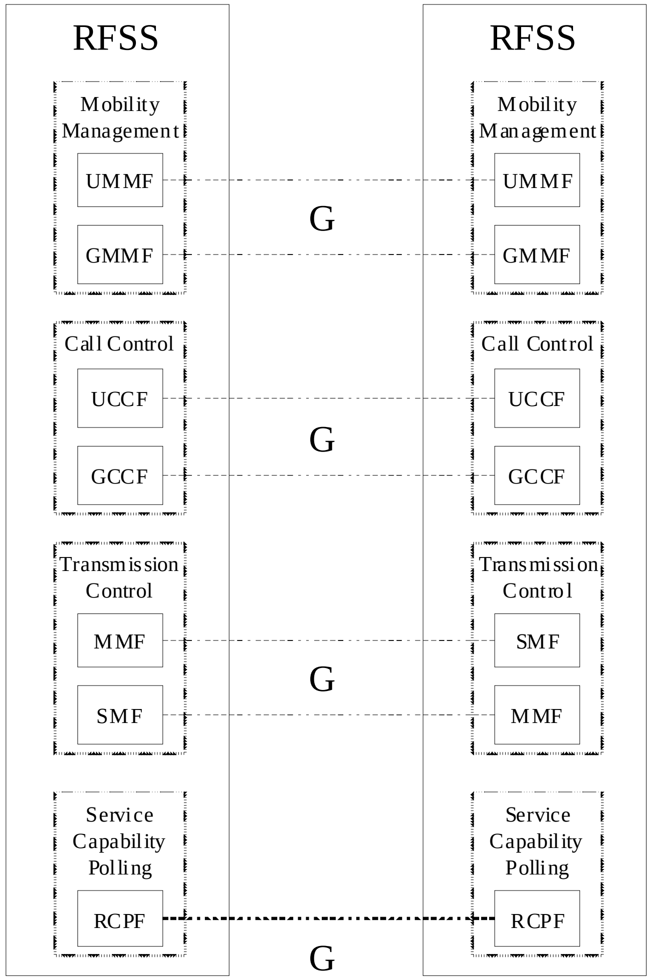

Figure 9 - ISSI Functional Entities

2.1.3.1 Mobility Management

*[No change from existing text in this section of **[102BACA]**.]*

2.1.3.2 Call Control

*[No change from existing text in this section of **[102BACA]**, except for Subsection 2.1.3.2.2 below.]*

2.1.3.2.1 Unit Call Control

*[No change from existing text in this section of **[102BACA]**.]*

2.1.3.2.2 Group Call Control

*[Replace existing text in this section of **[102BACA]** with the text below. Existing Subsections 2.1.3.2.2.1 – 2.1.3.2.2.4 remain unchanged but a new subsection 2.1.3.2.2.5 is added.]*

Group Call Control comprises the following services:

- Group call set up
- Group call segment set up, including determining the allowed vocoder mode for a call segment's talk spurts
- Group call segment termination
- Group call termination
- Confirmed Call Management
- Coordination of Group Emergency Cancellation

The **Group Call Control Function** (GCCF) is the functional entity activated in the RFSS which is in charge of the Call Control for the group call. The GCCF supports different functions depending on the role of the RFSS with respect to the group in question.

The GCCF of the Serving RFSS is responsible for the following functions:

- Requesting a group call set up when an SU triggers a group call request in its area for a group for which no call currently exists
- Informing the Home RFSS of its offered vocoder mode(s) when it requests that a group call be set up
- Participating in the establishment of new call segments in response to requests from the Home RFSS
- Releasing a group call segment with the Home RFSS on loss of RTP connectivity or other local policy condition
- Once a group call segment has been set up, tearing down or setting up RTP sessions upon preemption or Home RFSS request.
- Communicating to the Home RFSS the RTP session parameters
- Notifying the Home RFSS of the availability or lack of availability of RF resources required for a confirmed call
- Requesting a change of group call priority based on local policy
- Requesting a change of its call segment's allowed vocoder mode based on local policy
- Signaling a Group Emergency Cancellation to the Home RFSS when a group emergency is cancelled by a sufficiently privileged SU or due to local policy, such as when a Dispatch Console within the Serving RFSS cancels the group emergency.

The GCCF of the Home RFSS is responsible for the following functions:

- Establishing group call segments in response to group call setup requests from Serving RFSSs or SU PTT triggers within its own area, including:
  - to get the list of registered Serving RFSSs from the GMMF
  - Collaborating with the registered Serving RFSSs to determine the allowed vocoder mode for the talk spurts on each call segment
  - to initiate group call set ups with the registered Serving RFSSs
  - to bridge RTP sessions which are set up with the Serving RFSSs
  - to handle INVITE and re-INVITE request collisions
- On-going management of group call segments during a call, including
  - To initiate group call segment set ups with any registered Serving RFSSs not already joined to the call (late call entry)
  - To inform a registered Serving RFSS of a requested change of the allowed vocoder modes for the talk spurts on the call segment, based on the Serving RFSS's request and/or based on local policy
  - To handle a group call segment set up when it receives a group call set up request from any Serving RFSS
  - Once group call segment has been set up, tearing down or setting up RTP sessions upon preemption or Serving RFSS request.
- Releasing the group call on ISSI hangtime[2] expiry or other local policy condition
- Releasing a group call segment with a Serving RFSS on loss of RTP connectivity or other local policy condition
- Communicating to each Serving RFSS the RTP session parameters associated with its call segment
- Managing the establishment and maintenance of confirmed group calls
- Changing the group call priority upon Serving RFSS request or upon its own local policy.
- Signaling a Group Emergency Cancellation to the Serving RFSSs when it receives a Group Emergency Cancel from a sufficiently privileged SU or a Serving RFSS, or when it determines the group emergency to be over through its own local policy (e.g. when a Dispatch Console internal to the Home RFSS cancels the group emergency).

2.1.3.2.2.1 Group Call Topology

*[No change from existing text in this section of **[102BACA]**.]*

2.1.3.2.2.2 Group Call RTP Resource Management

*[No change from existing text in this section of **[102BACA]**.]*

2.1.3.2.2.3 Group Call RF Resource Management

*[No change from existing text in this section of **[102BACA]**.]*

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[2] Hangtime on the ISSI is not correlated to the RF resource hangtime. Generally, hangtime on the ISSI is longer than RF resource hangtime.

2.1.3.2.2.4 Group Call Vocoder Mode Selection
*[No change from existing text in this section of **[102BACA]**.]*

2.1.3.2.2.5 Group Emergency Signaling
*[This is a new section of **[102BACA]**.]*

When the GCCF in the Home RFSS receives a Group Emergency Cancel from a sufficiently privileged SU or from a Serving RFSS, the GCCF signals a Group Emergency Cancel to the Serving RFSSs. Also, the GCCF could signal a Group Emergency Cancel to the Serving RFSSs based on its local policy, such as when it receives a group emergency cancel from an internal wireline console.

When the GCCF in the Serving RFSS receives a Group Emergency Cancel from a sufficiently privileged SU, the GCCF signals a Group Emergency Cancel to the Home RFSS. Also, the GCCF could signal a Group Emergency Cancel to the Home RFSS based on its local policy, such as when it receives a group emergency cancel from an internal wireline console.

2.1.3.3 Media processing
*[No change from existing text in this section of **[102BACA]**.]*

2.1.3.4 RFSS Service Capability Polling
*[No change from existing text in this section of **[102BACA]**.]*

2.1.4 Service Access Point Model
*[No change from existing text in this section of **[102BACA]**, except for Subsection 2.1.4.2 below.]*

2.1.4.1 Mobility Management SAP
*[No change from existing text in this section of **[102BACA]**.]*

2.1.4.2 Call Control SAP
*[Replace existing text in this section of **[102BACA]** with the text below. Existing Subsections 2.1.4.2.1 – 2.1.4.2.3 remain unchanged but a new subsection 2.1.4.2.4 is added.]*

The Call Control SAP consists of four transactions, as summarized in Table 2 and illustrated in Figure 18. Detailed definitions of each of the supporting transactions and primitives are provided below.

Table 2 - Call Control SAP Functions

Transaction	Type	Purpose
CC_Setup	Rqst/Resp/Conf, Ind/IResp/IRConf	Provides the control function the ability to setup a call segment.
CC_Teardown	Rqst, Ind	Provides the control function with the ability to teardown a call segment.
CC_Modify	Rqst/Resp/Conf, Ind/IResp/IRConf	Provides the ability to modify a call segment.
CC_GrpEmerCancel	Rqst/Ind	Provides the ability to signal that a Group Emergency Cancel Request was received.

The CC_Setup transaction involves all six primitive types (Request, Response, Confirm, Indicate, Indicate Response, and IndicateResponseConfirm), as does the CC_Modify transaction. The CC_Teardown transaction only includes two primitive types, Request and Indicate. Likewise the, CC_GrpEmerCancel transaction only includes the Request and Indicate primitive types. Figure 2 illustrates the transactions based on these primitives.

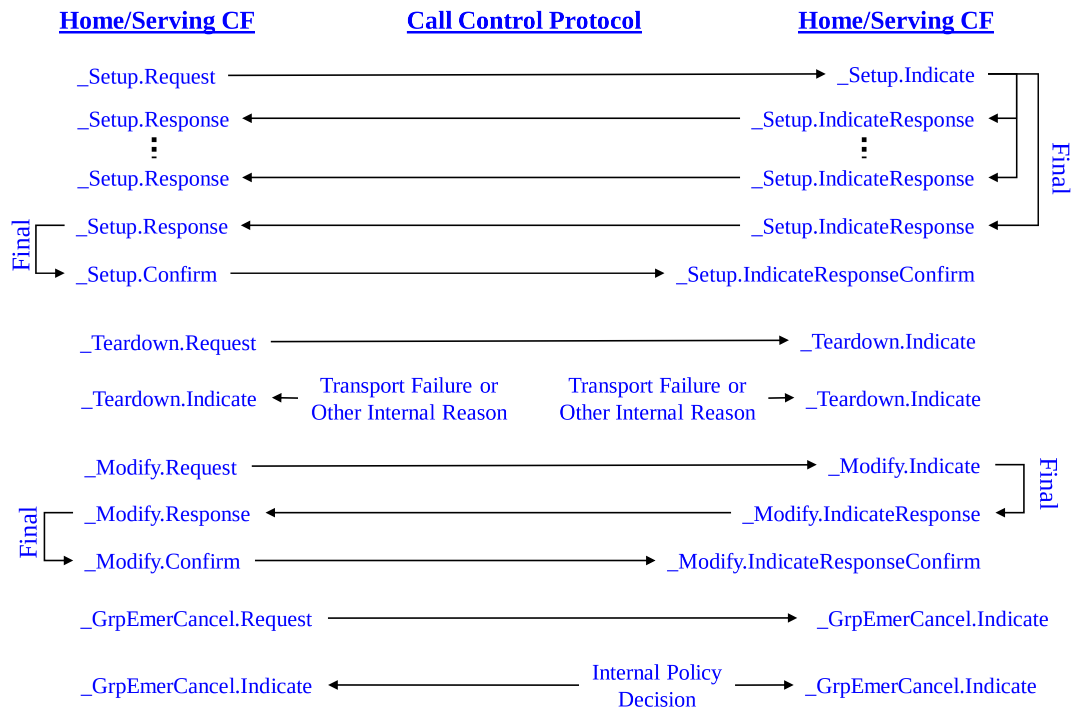

Figure 18 - CC SAP/Message Model

2.1.4.2.1 CC Call Segment Setup Transaction

[No change from existing text in this section of ***[102BACA]****.]*

2.1.4.2.2 CC Call Segment Teardown Transaction

[No change from existing text in this section of ***[102BACA]****.]*

2.1.4.2.3 CC Call Segment Modification Transaction

[No change from existing text in this section of ***[102BACA]****.]*

2.1.4.2.4 CC Group Emergency Cancel Transaction

[This is a new section of ***[102BACA]****.]*

The control function can use the *CC_GrpEmer_Cancel.Request* primitive to notify any peer RFSSs of the reception of a Group Emergency Cancel. Also, the *CC_GrpEmer_Cancel.Indicate* primitive informs the control function that a group emergency cancellation has been received from a peer RFSS or has been issued because of local policy.

2.1.4.3 Transmission Control SAP

*[No change from existing text in this section of **[102BACA]**.]*

2.2 ISSI Internet Protocol Suite

*[No change from existing text in this section of **[102BACA]**.]*

2.3 ISSI Data Transmission Order

*[No change from existing text in this section of **[102BACA]**.]*

3. SIP Messages and Parameters Definition for ISSI

*[No change from existing text in this section of **[102BACA]**, except for subsection 3.8 below.]*

3.1 Preamble: Grammar

*[No change from existing text in this section of **[102BACA]**.]*

3.2 SIP Transport

*[No change from existing text in this section of **[102BACA]**.]*

3.3 Request Line Construction

*[No change from existing text in this section of **[102BACA]**.]*

3.4 SIP URI and Related Addressing Logic

*[No change from existing text in this section of **[102BACA]**.]*

3.5 SIP Header Encoding

*[No change from existing text in this section of **[102BACA]**.]*

3.6 Status-Code and Reason-Phrase

*[No change from existing text in this section of **[102BACA]**.]*

3.7 SDP Announcements

*[No change from existing text in this section of **[102BACA]**.]*

3.8 X-TIA-P25-ISSI Body Parameters

*[No change from existing text in this section of **[102BACA]**, except for subsection 3.8.5 below.]*

3.8.1 Group Service Profile attributes

*[No change from existing text in this section of **[102BACA]**.]*

3.8.2 User Service Profile attributes

*[No change from existing text in this section of **[102BACA]**.]*

3.8.3 RFSS Service Capability Profile attributes

*[No change from existing text in this section of **[102BACA]**.]*

3.8.4 ISSI specific register parameter attributes

*[No change from existing text in this section of **[102BACA]**.]*

3.8.5 ISSI specific call parameter attributes

*[No change from existing text in this section of **[102BACA]**, except for subsection 3.8.5.6 below.]*

3.8.5.1 Initial transmitter

*[No change from existing text in this section of **[102BACA]**.]*

3.8.5.2 RF Resource Availability

*[No change from existing text in this section of **[102BACA]**.]*

3.8.5.3 Preference
*[No change from existing text in this section of **[102BACA]**.]*

3.8.5.4 Duplex Mode
*[No change from existing text in this section of **[102BACA]**.]*

3.8.5.5 In-call roaming
*[No change from existing text in this section of **[102BACA]**.]*

3.8.5.6 Group Call Type
*[Replace existing text in this section of **[102BACA]** with the text below.]*

This parameter indicates whether the group call is a confirmed or an unconfirmed group call.
It is named “c-groupcalltype ” with a compact form “c-gct”.

It has the following ABNF format:

group-call-type = (“c-groupcalltype:” / “c-gct:”) boolean CRLF

This field MAY be present only in an initial INVITE Request.

This field with value set to 1 (confirmed) SHALL be present in the INVITE Request from a Serving RFSS to a Home RFSS if the group service profile previously received from the Home RFSS contains the Confirmed Call Setup Time attribute with value greater than zero unless the SG has an active emergency condition. In that case, a Serving RFSS MAY disregard the group service profile parameter received from the Group Home RFSS and the field MAY be set to 0 (unconfirmed) to allow an emergency transmission to proceed unconfirmed on a confirmed SG.

This field with value set to 0 (unconfirmed) MAY be present in the INVITE Request from a Serving RFSS to a Home RFSS:

- If the group service profile previously received from the Home RFSS does not contain the Confirmed Call Setup Time attribute, or
- If the group service profile previously received from the Home RFSS contains the Confirmed Call Setup Time attribute with zero value.

For a confirmed group call, this field with value set to 1 (confirmed) SHALL be present in the INVITE Request from a Home RFSS to a Serving RFSS, and SHALL take precedence over the group service profile parameter. When an SG has an active emergency condition, a Home RFSS MAY set this field to 0 (unconfirmed) to allow an emergency transmission to proceed unconfirmed on a confirmed SG.

The default value is 0 (unconfirmed).

Figure 32 illustrates this usage.

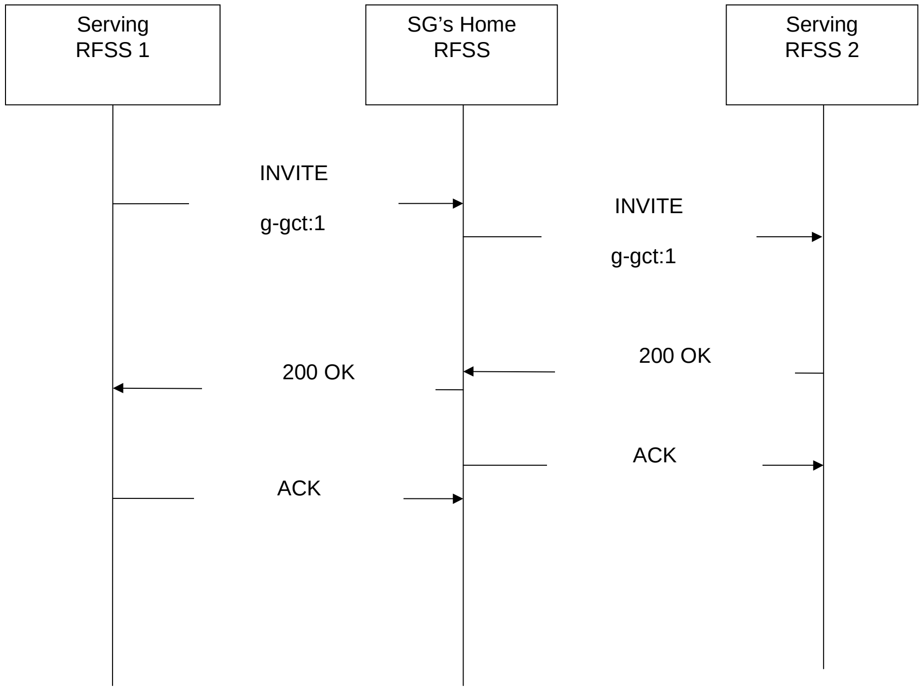

Figure 32 - Use of c-groupcalltype

3.8.5.7 Protected
*[No change from existing text in this section of **[102BACA]**.]*

3.8.5.8 Console Transmit Request Priority
*[No change from existing text in this section of **[102BACA]**.]*

3.9 X-TIA-P25-SNDCP Body Parameters
*[No change from existing text in this section of **[102BACA]**.]*

3.10 Examples
*[No change from existing text in this section of **[102BACA]**.]*

4. RTP Message Vocabulary for ISSI
*[No change from existing text in this section of **[102BACA]**.]*

5. Mobility Management
*[No change from existing text in this section of **[102BACA]**.]*

6. Call Control

*[No change from existing text in this section of **[102BACA]**, except for subsection 6.6 below.]*

6.1 CC-SAP

*[No change from existing text in this section of **[102BACA]**.]*

6.2 SIP Transactions

*[No change from existing text in this section of **[102BACA]**.]*

6.3 State Models for ISSI Voice Services

*[No change from existing text in this section of **[102BACA]**.]*

6.4 Basic ISSI SIP Control Procedures

*[No change from existing text in this section of **[102BACA]**.]*

6.5 Call Control Procedures for ISSI SU-to-SU Voice Services

*[No change from existing text in this section of **[102BACA]**.]*

6.6 Call Control Procedures for ISSI Group Voice Services

*[Replace existing text in this section of **[102BACA]** with the text below. As for its subsections, there is no change from existing text in **[102BACA]**, except for subsection 6.6.1 and the addition of new subsections 6.6.8 and 6.6.9.]*

Eight types of call control procedures are applicable to group voice services over the ISSI, as specified in the subsections below:

1. ISSI Group Voice Call Setup
2. Subsequent Modification of ISSI SIP Session Characteristics
3. Handling (re-)INVITE Request Collisions
4. ISSI SIP Heartbeat
5. ISSI SIP Session Termination by Home RFSS
6. ISSI SIP Session Termination by Serving RFSS
7. ISSI Group Voice Call Release
8. ISSI Group Emergency Cancellation

Only the first four procedure types involve SIP INVITE or re-INVITE requests.

In the event that a home RFSS, or a serving RFSS, receives a SIP INVITE or re-INVITE request pertaining to a call control procedure that it does not support, it SHALL respond with a GEN_PROC_UNSUPPORTED failure response as specified in Annex A of this document.

6.6.1 ISSI Group Voice Call Setup

[No change from existing text in this section of ***[102BACA]****.]*

6.6.1.1 General

[Replace existing text in this section of ***[102BACA]*** *with the text below.]*

A group call MAY be an unconfirmed or a confirmed group call.

For an unconfirmed group call, an RFSS initiates the group call over the ISSI upon receipt of an endpoint trigger (e.g., upon receipt of a group voice service request over the air interface as an end user pushes the PTT button on an SU affiliated with the group). The originating RFSS MAY be the Home RFSS or a Serving RFSS. If the originating RFSS is not the Home RFSS, the RFSS initiates the group call by sending a SIP INVITE request to the Home RFSS. If the Home RFSS accepts this SIP INVITE request, it would set up a call segment with the originating RFSS, and invite other Serving RFSSs in the group to join the call by sending individual SIP INVITE requests to them. If the originating RFSS is the Home RFSS, it directly invites other Serving RFSSs in the group to join the call. The Home RFSS bridges together individual RTP sessions (one for each call segment) and controls the transmission of RTP packets among the RFSSs in the group call. The unconfirmed group call is started over the ISSI as soon as the first SIP session and associated RTP session are successfully established between the Home RFSS and a Serving RFSS in the group, and talk spurt transmission over the ISSI MAY immediately begin.

For a confirmed group call, an RFSS initiates the group call over the ISSI upon receipt of an endpoint trigger, and the confirmed group call is started over the ISSI as soon as the first SIP session and associated RTP session are successfully established between the Home RFSS and a Serving RFSS in the group, similar to unconfirmed group call. However, talk spurt transmission does not begin until the Home RFSS determines that sufficient RF resources are available for the confirmed group call, or upon the expiry of a timer associated with the confirmed group call (refer to the procedures in the following subsections for details). Similar to unconfirmed calls, the originating RFSS MAY be the Home RFSS or a Serving RFSS.

When a confirmed group has an active emergency condition, the initiating RFSS MAY elect to initiate an unconfirmed group call in order to allow the emergency call to proceed without waiting for confirmed signaling.

For both unconfirmed and confirmed group calls, the procedures in the following subsections apply.

Upon receipt of an INVITE Request, the destination RFSS determines its role based on the values in the Route header field, the Request URI, and the To and/or the From header fields as specified in Section 6.4.1.1 of this document.

In a group call, the SIP messages within a SIP dialog SHALL have the same From and To fields. Each SIP dialog is distinguished by a Call-ID, To tag, and From tag. Various SIP dialogs in a group call MAY have different Call-IDs.

6.6.1.2 Group Voice Call Setup Initiated by Serving RFSS

*[Replace existing text in this section of **[102BACA]** with the text below, including all subsections.]*

In the event that the originating RFSS is the Home RFSS itself, procedures in Section 6.6.1.3 apply. Otherwise, the following procedures apply.

6.6.1.2.1 Actions Required at Serving RFSS

The Serving RFSS SHALL allocate RTP resources, and initiate a SIP INVITE request to the group's Home RFSS to ultimately establish an RTP session with the Home RFSS, according to the procedures in Section 6.4.1 of this document, with the following qualifications:

1. The SIP INVITE request SHALL include a Type 1 SDP offer (as defined in Section 3.7 of this document) that contains one or more media formats supported by the Serving RFSS, listed in the order of preference, with the first format listed being preferred, which MAY take into account information on the Home RFSS's set of supported SDP media formats that the Serving RFSS MAY have received during group registration over the ISSI (see Section 5).

2. The SIP INVITE request MAY include a "Group Call Type" parameter (as defined in Section 3.8.5.6) that indicates whether it is for a confirmed or unconfirmed group call. If a Serving RFSS elects to initiate an unconfirmed group call on a confirmed group with an active emergency condition, the SIP INVITE request SHALL include a "Group Call Type" parameter (as defined in Section 3.8.5.6).

3. The SIP INVITE request MAY include a "Protected" parameter (as defined in Section 3.8.5.7) that indicates whether the call's audio is requested to be protected.

4. For a confirmed group call, the SIP INVITE request SHALL include an "RF Resource Availability" parameter (as defined in Section 3.8.5.2) that indicates whether the Serving RFSS has available RF resource to support the confirmed group call. In the event that RF resource is currently unavailable, the Serving RFSS SHALL send a re-INVITE request, with appropriate "RF Resource Availability" parameter setting, to the Home

RFSS when RF resource is later available, following the procedures in Section 6.4.3 of this document.

5. The Serving RFSS MAY cancel its SIP INVITE request, following procedures in Section 6.4.2 of this document.

6. Upon receipt of a VCE_SG_NOT_REG or VCE_SG_UNKNOWN failure response, the Serving RFSS SHALL consider itself not registered to the group and SHOULD, before attempting a new INVITE request, initiate registration with the Home RFSS, following procedures in Section 5 of this document.

7. In the event that, while awaiting a response on the SIP INVITE request, the Serving RFSS receives a SIP INVITE or re-INVITE request from the Home RFSS, the Serving RFSS SHALL handle the received SIP INVITE or re-INVITE request, according to procedures in Section 6.6.3 of this document.

8. In the event that the “Protected” parameter, as specified in Section 3.8.5.7 of this document, is present in the received SIP 200 OK ISSI-specific body, or if the parameter is not present and takes on the implied default value of “unprotected”, then the Serving RFSS SHALL handle the call initially according to the received “Protected” value (be it protected or unprotected).

9. In the event that a SIP 3xx, 4xx, 5xx, or 6xx response for the SIP INVITE is received, the Serving RFSS MAY retry the SIP INVITE request subject to local policy.

6.6.1.2.2 Actions Required at Home RFSS

The Home RFSS SHALL process the received SIP INVITE request, according to procedures in Section 6.4.1 of this document, with the following qualifications:

1. In the event that the Serving RFSS initiating the SIP INVITE request is not registered to the group, the Home RFSS SHALL return a VCE_SG_NOT_REG failure response.

2. In the event that the group is not known to the Home RFSS, the Home RFSS SHALL return a VCE_SG_UNKNOWN failure response response.

3. In the event that the Home RFSS detects that none of the SDP media formats offered by the Serving RFSS is accepted by the Home RFSS, it SHALL return a VCE_VOCODER_MODES_UNACCEPTED failure response.

NOTE: For older RFSS implementations that only support the Full Rate vocoder mode with encoding-name of the rtpmap attribute set to “X-TIA-P25-IMBE” (see

Section 3.7), a GEN_SDP_PARAMS_UNACCEPTABLE failure response is returned instead.

4. The Home RFSS SHALL process any received SIP CANCEL or BYE request following the procedures in Section 6.4.2.1 or 6.4.4.1 of this document.

5. In the event that the Home RFSS is able to allocate RTP resource to set up the RTP session with the originating RFSS, it SHALL return a SIP 200 (OK) response that carries a Type 1 SDP answer, and it SHALL proceed to invite other Serving RFSSs registered to the Home RFSS to join the call, according to the procedures in the following subsections, with the following qualification:

    a. The Type 1 SDP answer in the SIP 200 (OK) response SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Serving RFSS in its SIP INVITE request, has been accepted by the Home RFSS, which MAY take into account information on the sets of SDP media formats supported by other Serving RFSSs in the same group that the Home RFSS MAY have received during their group registrations over ISSI (see Section 5).

6. In the event that the Home RFSS is unable to allocate RTP resource, it SHOULD return a SIP 200 (OK) response that carries a Type 2 SDP answer, and SHOULD send a re-INVITE request that carries a Type 1 SDP offer to the Serving RFSS when RTP resource is later available, following the procedures in Section 6.4.3 of this document. In this situation, and if the group call has not been already set-up with another Serving RFSS, the Home RFSS SHOULD NOT proceed to invite other Serving RFSSs registered with the Home RFSS for that group. The following qualification apply:

    a. The Type 2 SDP answer in the SIP 200 (OK) response SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Serving RFSS in its SIP INVITE request, has been accepted by the Home RFSS, which MAY take into account information on the sets of SDP media formats supported by other Serving RFSSs in the same group that the Home RFSS MAY have received during their group registrations over ISSI (see Section 5).

7. In the event that the “Protected” parameter, as specified in Section 3.8.5.7 of this document, is present in the received INVITE request’s ISSI-specific body, or if the parameter is not present and takes on the implied default

value of “unprotected”, the Home RFSS SHALL determine whether the SG supports the value of this parameter:

a. In the event that the SG supports this parameter value, then the Home RFSS SHALL include the parameter value in its 200 OK response to the Serving RFSS, and in the SIP INVITE requests to the other Serving RFSSs for the SG. (see Section 6.6.1.3).

b. In the event that the SG does not support this parameter value, then the Home RFSS SHALL return a VCE_CANNOT_PROTECT_AUDIO or VCE_CANNOT_UNPROTECT_AUDIO failure response, as appropriate, to the Serving RFSS.

8. In the event that the “Group Call type” parameter, as specified in Section 3.8.5.6 of this document, is present in the received INVITE request’s ISSI-specific body and it takes on the value of “unconfirmed” for a confirmed group with an active emergency condition, the Home RFSS MAY proceed to process the call as an unconfirmed call.

    a. If the Home RFSS proceeds to process the call as a confirmed call, the requesting Serving RFSS SHALL handle the reception of a subsequent PTT Transmit Wait packet as defined in 7.9.5.

6.6.1.3 Call Segment Setup Initiated by Home RFSS

*[Replace existing text in this section of **[102BACA]** with the text below, including all subsections.]*

The Home RFSS MAY initiate setup of one or more call segments with the Serving RFSSs that are already registered to the group, under the following situations:

1. The Home RFSS initiates the group call and therefore directly invites Serving RFSSs in the group to join the call.

2. A Serving RFSS initiates the group call, and, upon successful setup of a call segment between the Home and Serving RFSS, the Home RFSS invites other Serving RFSSs in the group to join the call.

6.6.1.3.1 Actions Required at Home RFSS

The Home RFSS SHALL allocate RTP resources for RTP sessions with other Serving RFSSs within the group that are successfully registered with the Home RFSS, but have not yet been invited to join the group call. Throughout the group call, the Home RFSS SHOULD, with allocated RTP resources, initiate SIP INVITE or re-INVITE requests to any Serving RFSS in the group that is not already participating in the call.

The Home RFSS SHALL then initiate SIP INVITE requests to these Serving RFSSs, to ultimately establish one-to-one RTP sessions with them, following the procedures in Section 6.4.1 of this document, with the following qualifications:

1. The SIP INVITE request SHALL include a Type 1 SDP offer (as defined in Section 3.7 of this document) that contains one or more media formats supported by the Home RFSS, listed in the order of preference, with the first format listed being preferred, taking into account SDP media format(s) that the Home RFSS MAY already be using for talk spurt transmissions with some other Serving RFSSs in the group. The Type 1 SDP offer MAY also take into account information on the sets of SDP media formats supported by this and other Serving RFSSs in the same group that the Home RFSS MAY have received during their group registrations over ISSI (see Section 5).

2. The SIP INVITE request MAY include a "Group Call Type" parameter (as defined in Section 3.8.5.6) that indicates whether it is for a confirmed or unconfirmed group call. If a Home RFSS elects to initiate an unconfirmed group call on a confirmed group with an active emergency condition, the SIP INVITE request SHALL include a "Group Call Type" parameter (as defined in Section 3.8.5.6).

3. The SIP INVITE request MAY include a "Protected" parameter (as defined in Section 3.8.5.7) that indicates whether the call's audio is requested to be protected. (see Section 6.6.1.2.2)

4. In the event that, while awaiting a final response on the SIP INVITE request to a given Serving RFSS, the Home RFSS receives a SIP INVITE or re-INVITE request from the Serving RFSS, the Home RFSS SHALL handle the received SIP INVITE or re-INVITE request, according to procedures in Section 6.6.3 of this document. In the event that the Home RFSS originated SIP INVITE request does not have precedence, the Home RFSS SHALL wait for a SIP final response from each remaining Serving RFSS in the group (i.e., any Serving RFSS in the group except the Serving RFSS whose SIP INVITE request has precedence) that has not yet returned any final response to the SIP INVITE from the Home RFSS, following the procedures in Section 6.4.1.1 of this document. Subsequently, the Home RFSS:

a. SHOULD send a SIP re-INVITE request to each remaining Serving RFSS in the group that has returned a 2xx final response to the SIP INVITE from the Home RFSS, following the procedures in Section 6.4.3.1 of this document, to update the session parameters to match those of the new call initiated by the given Serving RFSS.

b. SHALL initiate SIP INVITE requests to those Serving RFSSs in the group that have not yet joined the group call, following the procedures in Section 6.6.1.3 of this document.

5. Any time after receiving a SIP 1xx provisional response and before receiving a final response from the Serving RFSS, the Home RFSS MAY cancel its SIP INVITE request, following procedures in Section 6.4.2 of this document.

6. Upon receipt of a VCE_SG_UNKNOWN failure response, the Home RFSS SHALL consider the Serving RFSS no longer registered to the group and SHOULD remove this RFSS from its list of Serving RFSSs for the group.

7. In the event that a SIP 3xx, 4xx, 5xx, or 6xx response for the SIP INVITE is received, the Home RFSS MAY retry the SIP INVITE request subject to local policy.

For an unconfirmed group call, talk spurt transmission MAY start as soon as the first SIP session and associated RTP session for a group call have been successfully established between the Home RFSS and a Serving RFSS.

For a confirmed group call, after successful setup of the first SIP session and RTP session, the Home RFSS SHALL control talk spurt transmission according to the following procedures:

1. Upon receipt of the first PTT Transmit Request from a Serving RFSS, the Home RFSS SHALL check whether sufficient number of RFSSs have RF resources available to support the confirmed group call.

2. In the event that sufficient RF resources are available, the Home RFSS SHALL grant the first PTT Transmit Request the permission to transmit talk spurts.

3. In the event that insufficient RF resources are available, the Home RFSS SHALL start a Tgchstartconfirmed timer.

a. In the event that sufficient RF resources later become available, the Home RFSS SHALL stop the Tgchstartconfirmed timer and grant one of the received PTT Transmit Requests the permission to transmit, using the selection criteria as specified in Section 7.9.1 of this document.

b. Upon expiry of the Tgchstartconfirmed timer, the Home RFSS SHOULD grant one of the received PTT Transmit Requests the permission to transmit talk spurts, using the selection criteria as specified in Section 7.9.1 of this document.

4. Depending on local policy, the Home RFSS MAY or MAY NOT check RF resource availability , by repeating the above steps, before granting each subsequent PTT Transmit Request the permission to transmit.

Moreover, as soon as the first SIP session for a confirmed or unconfirmed group call have been successfully established between the Home RFSS and a Serving RFSS, the Home RFSS SHALL start a Tgchhangtime timer that is reinitialized every time the Home RFSS detects voice activity on the RTP sessions for the group call.

As each RTP session is established, the Home RFSS SHALL bridge it to other RTP session(s) established earlier, and SHALL control the ensuing talk spurts among the various RFSSs participating in the group call, following the procedures in Section 7 of this document.

6.6.1.3.2 Actions Required at Serving RFSS

The Serving RFSS SHALL process the received SIP INVITE request, according to the procedures in Section 6.4.1 of this document, with the following qualifications:

1. In the event that the group is not known to the Serving RFSS, the Serving RFSS SHALL return a VCE_SG_UNKNOWN failure response.

2. In the event that the Serving RFSS detects that none of the SDP media formats offered by the Home RFSS is accepted by the Serving RFSS, it SHALL return a VCE_VOCODER_MODES_UNACCEPTED failure response.

NOTE: For older RFSS implementations that only support the Full Rate vocoder mode with encoding-name of the rtpmap attribute set to “X-TIA-P25-IMBE” (see Section 3.7), a GEN_SDP_PARAMS_UNACCEPTABLE failure response is returned instead.

3. In the event that the Serving RFSS is able to allocate RTP resource to set up the RTP session with the Home RFSS, it SHALL return a SIP 200 (OK) response that carries a Type 1 SDP answer. The Type 1 SDP answer SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Home RFSS in its SIP INVITE request, has been accepted by the Serving RFSS.

4. In the event that the Serving RFSS is unable to allocate RTP resource, it SHOULD return a SIP 200 (OK) response that carries a Type 2 SDP answer, and SHOULD send a re-INVITE request that carries a Type 1 SDP offer to the Home RFSS when RTP resource is later available, following the procedures in Section 6.4.3 of this document. The following qualifications apply:

    a. The Type 2 SDP answer SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Home RFSS in its SIP INVITE request, has been accepted by the Serving RFSS.

    b. The subsequent Type 1 SDP offer SHALL indicate which particular media format, chosen from the set of SDP media formats originally offered by the Home RFSS in its SIP INVITE request, is now offered by the Serving RFSS, also taking into account the media format in the Type 2 SDP answer that the Serving RFSS previously sent to the Home RFSS.

5. For a confirmed group call, in the event that the 200 (OK) conveys a Type 1 SDP answer, it SHALL include an “RF Resource Availability” parameter (as defined in Section 3.8.5.2) that indicates whether the Serving RFSS has available RF resource to support the confirmed group call. The Type 1 SDP answer SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Home RFSS in its SIP INVITE request, has been accepted by the Serving RFSS. In the event that RF resource is currently unavailable, the Serving RFSS SHALL send a re-INVITE request, with appropriate “RF Resource Availability” parameter setting, to the Home RFSS when RF resource is later available, following the procedures in Section 6.4.3 of this document.

6. In the event that the “Protected” parameter, as specified in Section 3.8.5.7 of this document, is present in the received INVITE request’s ISSI-specific body, or if the parameter is not present and takes on the implied default value of “unprotected”, then the Serving RFSS SHALL handle the call initially according to the received “Protected” value (be it protected or unprotected).

7. The Serving RFSS SHALL process any received SIP CANCEL or BYE request following the procedures in Sections 6.4.2.1 and 6.4.4.1 of this document.

6.6.2 Subsequent Modification of ISSI SIP Session Characteristics

*[No change from existing text in this section of **[102BACA]**.]*

6.6.3 Handling (re-)INVITE Request Collisions

*[No change from existing text in this section of **[102BACA]**.]*

6.6.4 ISSI SIP Heartbeat

*[No change from existing text in this section of **[102BACA]**.]*

6.6.5 ISSI SIP Session Termination by Home RFSS

*[No change from existing text in this section of **[102BACA]**.]*

6.6.6 ISSI SIP Session Termination by Serving RFSS

*[No change from existing text in this section of **[102BACA]**.]*

6.6.7 ISSI Group Voice Call Release

*[No change from existing text in this section of **[102BACA]**.]*

6.6.8 ISSI Group Emergency Cancellation

*[This is a new section of **[102BACA]**.]*

6.6.8.1 Group Emergency Cancellation Initiated by Serving RFSS

A Serving RFSS SHALL send a Group Emergency Cancel message per **[102BACD]** to the Group Home RFSS when it receives a group emergency cancellation for an SG from a sufficiently privileged SU.

A Serving RFSS MAY send a Group Emergency Cancel message per **[102BACD]** to the Group Home RFSS when the Serving RFSS determines that the Group Emergency condition is over based on its own local policy, such as when the RFSS receives a group emergency cancellation from a wireline dispatch console within the RFSS.

A Serving RFSS that is a Tracking RFSS SHALL cancel the emergency state of the SG upon sending the Group Emergency Cancel message to the Group Home RFSS.

6.6.8.2 Group Emergency Cancellation Initiated by Home RFSS

The Group Home RFSS SHALL send a Group Emergency Cancel message per **[102BACD]** to the Serving RFSSs:

1) When the Group Home RFSS receives a Group Emergency Cancellation from a Serving RFSS. – OR–

2) When the Group Home RFSS receives a Group Emergency Cancellation from a sufficiently privileged SU.

The Group Home RFSS MAY send a Group Emergency Cancel message per **[102BACD]** to the Serving RFSSs when the Group Home RFSS determines that the Group Emergency condition is over based on its own local policy, such as when the RFSS receives a group emergency cancellation from a wireline dispatch console within the RFSS.

A Serving RFSS that is a Tracking RFSS SHALL cancel the emergency state of the SG when it receives a Group Emergency Cancel from the Home RFSS per **[102BACD]**. A Serving RFSS that is a Non-Tracking RFSS need not act on the Group Emergency Cancel message.

7. Push-to-Talk (PTT) Management

[No change from existing text in this section of ***[102BACA]****.]*

8. RFSS Service Capability Polling

[No change from existing text in this section of ***[102BACA]****.]*

Annex A (Normative) – Generic SIP Responses Over the ISSI

*[No change from existing text in this section of **[102BACA]**.]*

Annex B (Normative) – Table of Timers and Constants

*[No change from existing text in this section of **[102BACA]**.]*

Annex C (Normative) – TIA-102 Vocoder Error Mitigation Data

*[No change from existing text in this section of **[102BACA]**.]*

Annex D (Normative) – SU-to-SU Call Flows

*[No change from existing text in this section of **[102BACA]**.]*

Annex E (Normative) – Group Call Flows

*[No change from existing text in this section of **[102BACA]**.]*

Annex F (Informative) – SU-to-SU Call Flows Involving Multiple Vocoder Modes

*[No change from existing text in this section of **[102BACA]**.]*

Annex G (Informative) – Group Call Flows Involving Multiple Vocoder Modes

*[No change from existing text in this section of **[102BACA]**.]*

Annex H (Informative) – Future Considerations

*[No change from existing text in this section of **[102BACA]**.]*

Annex I (Informative) – Console Functionality

*[No change from existing text in this section of **[102BACA]**.]*

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TELECOMMUNICATIONS INDUSTRY ASSOCIATION

1320 N. Courthouse Road, Suite 200
Arlington VA, 22201, USA

+1.703.907.7700 MAIN
+1.703.907.7727 FAX